

Economic History II

Bureaucracy and public service delivery

FGV EPGE

Outline of the class

- **Three papers**, taken separately. All on bureaucrats and what the state delivers.
- Aneja and Xu (2024): civil service reform in the US post office during the Gilded Age. *American Economic Review*.
- Barteska and Lee (2026): the bureaucrats behind South Korea's export miracle. Working paper.
- Akerman, Moscona, Pellegrina and Sastry (2025): Embrapa and Brazil's agricultural revolution. Working paper.

Post offices, patronage and reform

Aneja and Xu (2024), *Strengthening State Capacity: Civil Service Reform and Public Sector Performance during the Gilded Age*, AER

The question

- An impartial civil service with lifelong careers: the standard prescription for a working state, since Weber.
- Direct evidence that such reforms improve state effectiveness: scarce.
- The setting is the **Pendleton Act of 1883**, the first nationwide civil service reform in the United States.
- The organization is the post office, the largest civilian agency of the federal government.

Strengthening State Capacity: Civil Service Reform and Public Sector Performance during the Gilded Age[†]

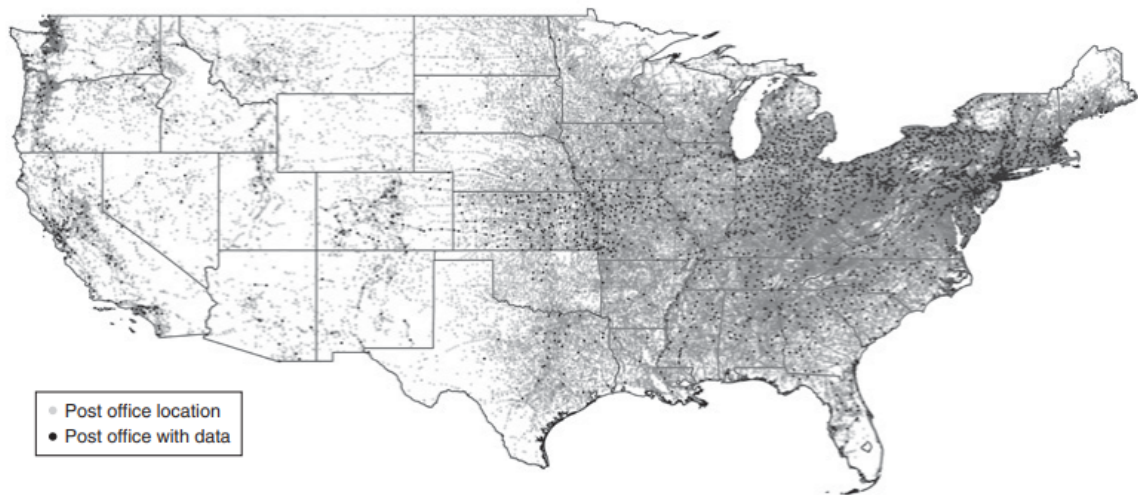
By ABHAY ANEJA AND GUO XU*

We use newly digitized records from the post office to study the effects of strengthened state capacity between 1875 and 1901. Exploiting the implementation of the Pendleton Act—a landmark statute that shielded bureaucrats from political interference—across US cities over two waves, we find that civil service reform reduced postal delivery errors and increased productivity. These improvements were most pronounced during election years when the reform dampened bureaucratic turnover. We provide suggestive evidence that reformed cities witnessed declining local partisan newspapers. Separating politics from administration, therefore, not only improved state effectiveness but also weakened the role of local politics. (JEL D24, D73, H83, J45, L82, L87, N41)

The post office in the nineteenth century

- At the onset of the Pendleton Act, postal workers were **34 percent of the entire federal workforce**.
- Principal method of long-distance communication in an expanding nation.
- Offices in virtually all parts of the country, all performing the same function: collection and delivery of mail, by clerks and carriers.
- That uniformity is what makes performance comparable across cities and years.
- John (1995): for most Americans, “the postal system was the central government”.

Panel A. All post offices



The spoils system

- Before 1883: patronage-based hiring and firing, with elected politicians holding discretion over appointments.
- Patronage employees campaigned for their benefactors and paid part of their salary back as **political assessments**.
- Eaton (1885) on what the parties extracted: “the plunder of the salaries of the public servants for filling the party treasury”.
- Careers tied to particular politicians, so turnover was high.
- One report on the New York City Post Office: “incompetency, neglect, confusion and drunkenness”.

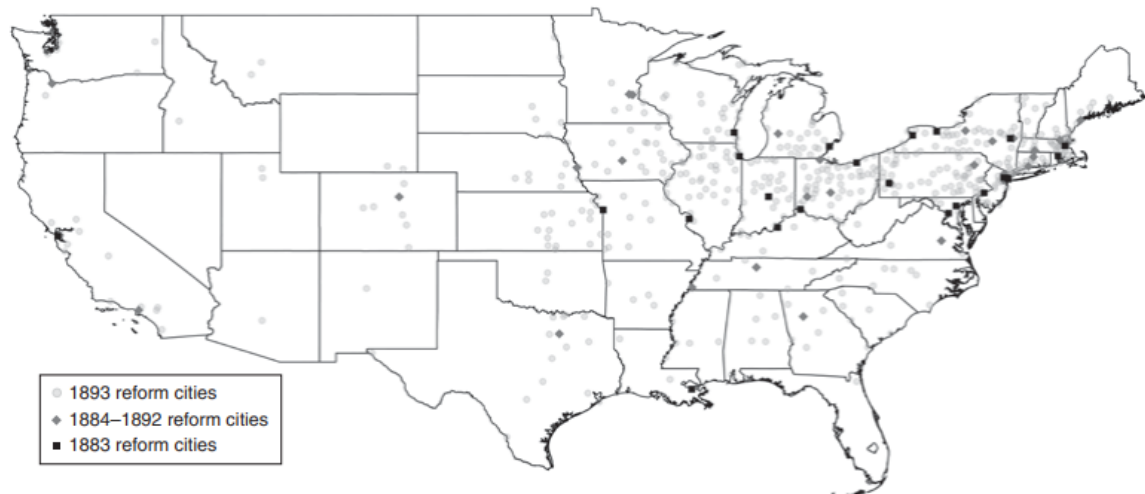
The Pendleton Act (1883)

- Discretionary appointments replaced by rule-based hiring through open, competitive **examinations**.
- Classified workers protected from removal without cause.
- Assessments outlawed for the whole federal government.
- Catalyzed by the assassination of President Garfield in 1881 by a disappointed office seeker.
- About 10 percent of federal employees were classified at the outset.

The reform arrived in waves

- Coverage was decided in Washington and keyed to size, not to local preference.
- 1883: post offices with at least 50 employees. **23 cities.**
- 1893: Harrison's executive order classified every post office with free delivery service.
556 more cities.
- In between, 30 cities crossed the 50-employee threshold on their own.
- By 1921, 80 percent of the federal civilian workforce was covered.

Panel B. Expansion of civil service reform



The data

- Large-scale digitization of the post office's own annual reports.
- **Delivery errors:** errors on returned slips, city by year, 1879 to 1901, 2,319 cities.
- Free delivery statistics: volume per carrier and cost per volume, 512 large cities, 1875 to 1891.
- Official Registers: every clerk and carrier, 1877 to 1901, 297,932 individual-year observations, linked to the full-count censuses of 1880 and 1900 (36 percent match).
- Newspaper directories: partisan and independent papers by city and election year.

C Census linking

We match the personnel records from the Official Registers of the United States (the “Registers”) to the U.S. full count Decennial Census to obtain additional individual background characteristics. Each postal worker is matched based on the full name (first name, middle name/initial, last name), birth state, and current state of employment. We proceed by using different combinations, successively relaxing the matching restrictions:

- Step 1: First name + middle name + last name + birth state + current state
- Step 2: First name + middle name initial + last name + birth state + current state
- Step 3: First name + last name + birth state + current state
- Step 4: First name + last name + birth state

Given the limited number of variables that are available for matching, we pursue a conservative approach to ensure we do not overmatch by linking incorrect individuals (i.e., false positives). To start with, we always match individual names exactly. Second, we discard candidate matches of Census respondents who were younger than

Table C1: Characteristics of Census-linked vs. non-linked workers

	(1)	(2)	(3)	(4)
	<u>Mean characteristics</u>		<u>Differences</u>	
	Matched	Unmatched	Raw	Conditional
<i>Panel A: Individual characteristics</i>				
Log(Salary)	5.80	5.83	-0.025 (0.007)	0.016 (0.007)
Clerk	0.59	0.60	-0.011 (0.002)	-0.006 (0.002)
Same state	0.52	0.66	-0.141 (0.002)	-0.147 (0.002)
Foreign-born	0.11	0.10	0.008 (0.001)	0.011 (0.002)
German	0.02	0.02	0.000 (0.000)	0.002 (0.001)
Irish	0.022	0.021	0.001 (0.001)	0.002 (0.001)

Where the error counts come from

- Clerks bundled mail by destination and attached a **facing slip** naming the destination and the sending clerk.
- Railway Mail Service clerks sorted the mail in transit and recorded every misdirected item.
- Errors were double-checked by a second examiner and reported back to supervisors.
- The Railway Mail Service was an independent arm of the post office, outside the city-level rollout.

How the effect is measured

- Compare cities **before and after** the reform reaches them with cities not yet reformed.
- The rollout rule was federal and size-based, so the timing was not a local political choice.
- Errors were recorded by a body with no stake in the local office, which limits strategic reporting.
- Effects traced year by year around the arrival of the reform, and separately for each wave.

Results

Fewer delivery errors

- Reformed cities: delivery errors fall by **19 to 22 percent**.
- Holds controlling for the size rule that assigned treatment, and for a wide set of baseline city characteristics.
- No difference in trend beforehand. The gap opens at the reform, widens, then levels out.
- The same dynamic pattern in 1883 and in 1893, a decade apart.
- Not driven by any single city or state.

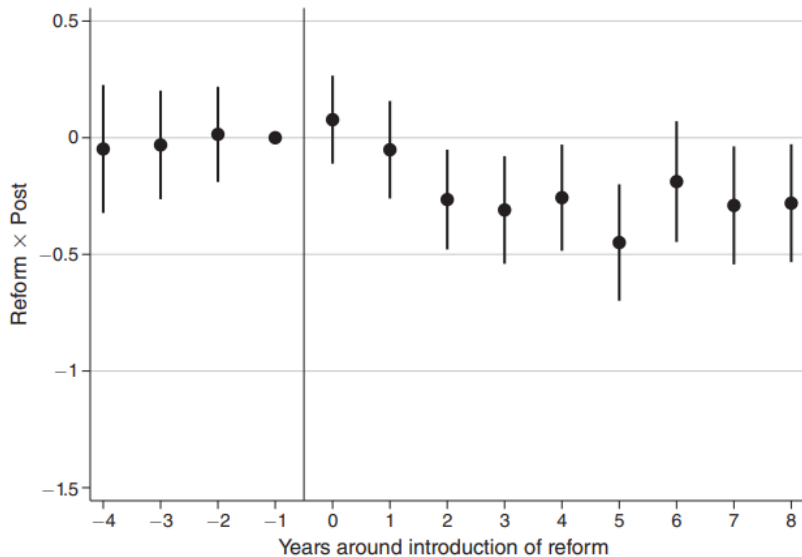
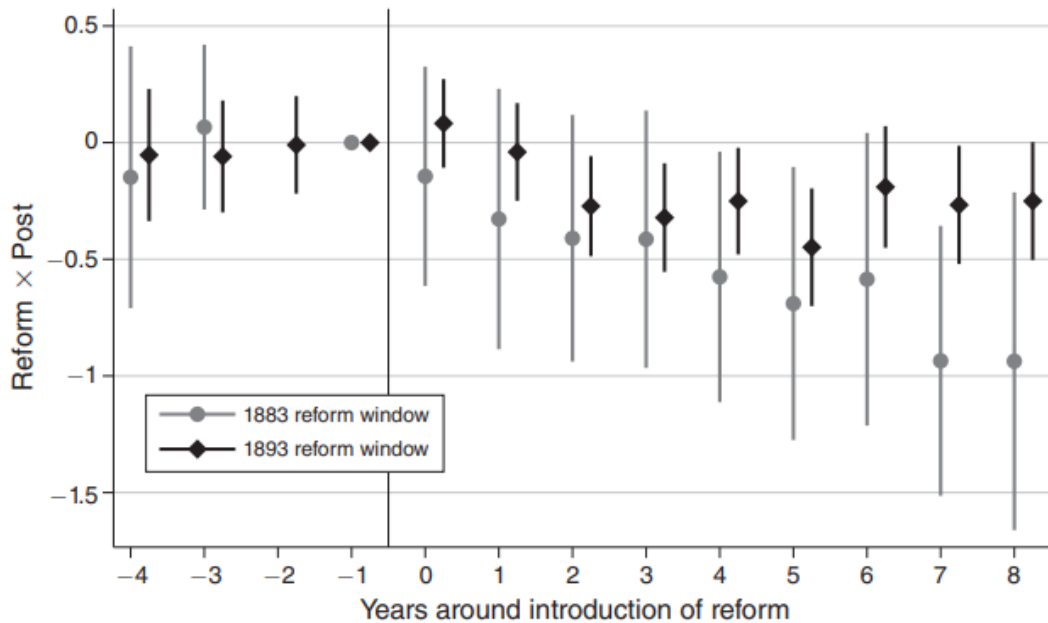


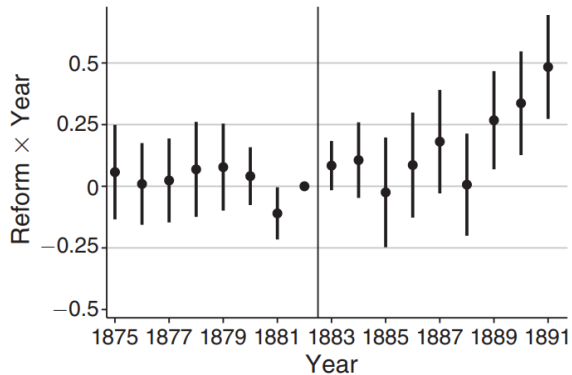
FIGURE 2. DELIVERY ERRORS FOR REFORM VERSUS NONREFORM CITIES AROUND REFORM YEARS



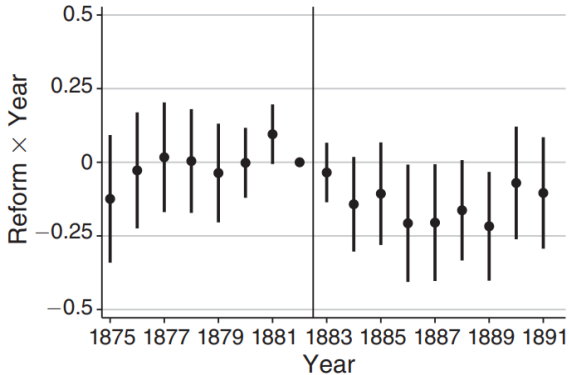
Higher productivity

- A second, independent record series: the City Free Delivery statistics, for the 1883 wave.
- Mail volume per carrier rises **8 to 14 percent**.
- Cost per volume falls 10 to 13 percent.
- Two measures from two sources, pointing the same way.

Panel A. Mail per carrier



Panel B. Cost per volume



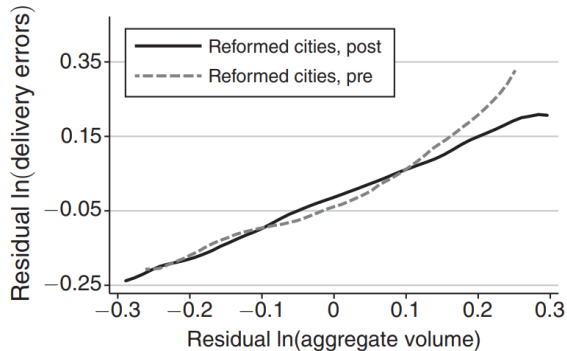
Ruling out the simple explanations

- More staff? Total postal employment moves 1.4 to 1.7 percent, indistinguishable from zero. Errors per worker fall as well.
- Different recruits? Home state, foreign birth, nationality, age, literacy, race and pay of new hires are all unchanged.
- Less work? Mail volume, if anything, rose.
- What changes is the **link between volume and errors**.

Volume and errors

- Before the reform, delivery errors rose with the volume of mail handled.
- After the reform, that relationship flattens in reformed cities.
- No comparable change among unreformed cities over the same years.
- The reform altered the “technology” of production rather than the workload.

Panel A. Reformed cities



Panel B. Unreformed cities

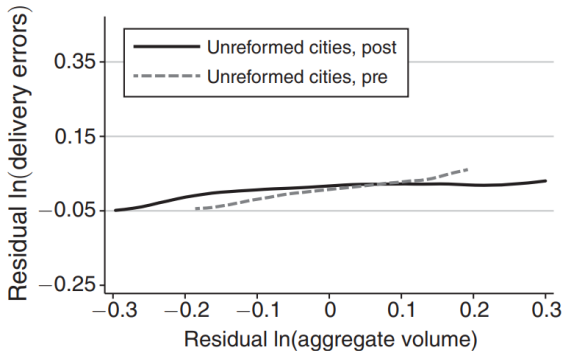
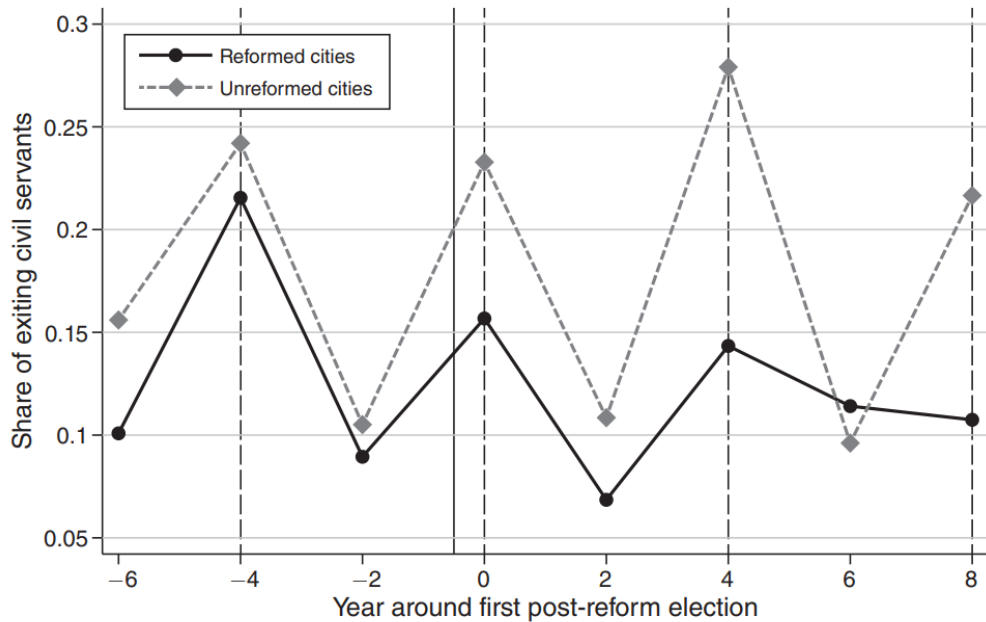


FIGURE 5. DELIVERY ERRORS AND AGGREGATE MAIL VOLUME AROUND THE REFORM

Turnover and the electoral cycle

- Serving at the pleasure of politicians meant insecure tenure. Hoogenboom (1959): “morale was low in a civil service largely composed of misfits employed on a temporary basis”.
- Annual exit rate among postal workers: **17 percent**, spiking in the years after presidential elections.
- The reform cuts the exit rate by about 4 points, a relative decline of 21 to 23 percent.
- Mean years of service rise by 0.8 to 1.0, from a base of 1.7.
- The electoral spike in exits flattens once a city is reformed.



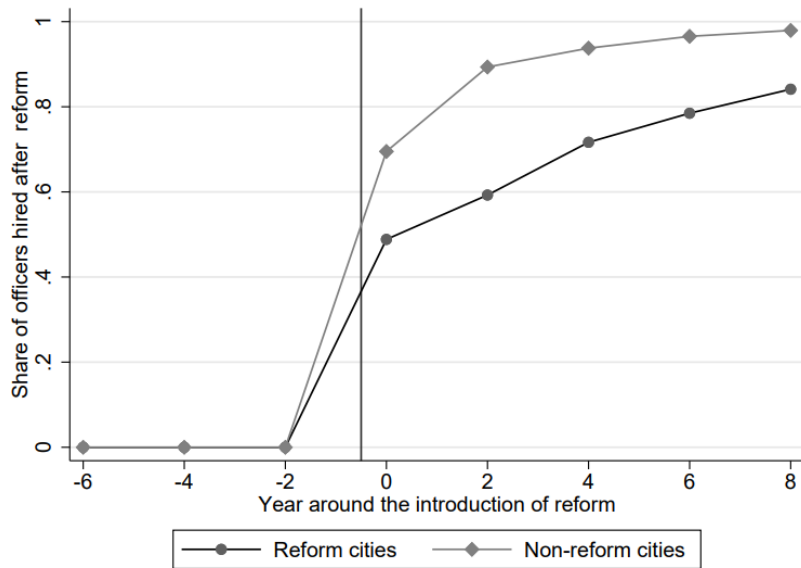
The gains land in election years

- The fall in delivery errors is driven by **election years**.
- So are the gains in volume per carrier and in cost per volume.
- Those were the years when patronage turnover previously destroyed accumulated experience.
- Retention improves for everyone serving in a reformed office, whatever their year of hire.

Selection was too slow to be the story

- Competitive exams applied only to new entrants, so replacing the existing staff was a protracted process.
- Eight years after the rules took effect, close to **20 percent** of the average reformed office were still patronage appointees.
- Lower exit is not confined to the post-reform entry cohort: it holds for those appointed under patronage too.
- The operating margin was the limit on arbitrary firing.

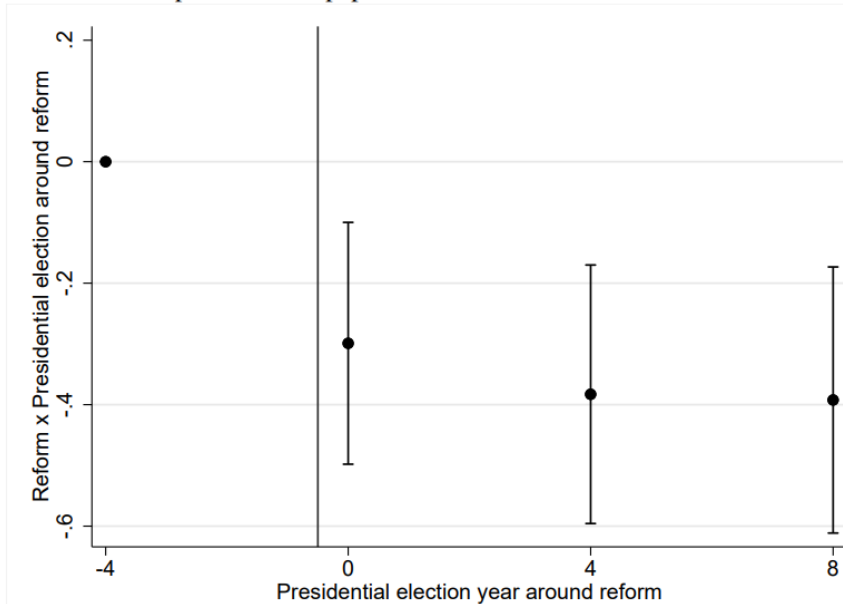
Figure A4: Share of civil servants hired after the reform event, by reform status



A side effect: weaker local parties

- Appointments were formally federal, in practice delegated to local politicians, and a source of political rents.
- Fewer patronage jobs meant less to trade for local political and financial support.
- Proxy for the strength of local parties: explicitly **partisan newspapers**, whose editors were often party officers.
- Partisan papers per city fall by 0.2 against a mean of 1.8, an 11 percent decline. Independent papers rise.
- The partisan share of circulation falls by about 4 points.

Figure A10: Number of political newspapers in reform vs. non-reform cities around reform year



In short: Aneja and Xu

- The Pendleton Act improved delivery and productivity in the largest civilian agency of the federal state.
- The channel was severing the tie between politicians and rank and file bureaucrats.
- Stable careers rather than a different pool of recruits.
- The authors call it a “double dividend”: a more effective bureaucracy, and weaker local parties.

The bureaucrats of the Korean miracle

Barteska and Lee (2026), *The Implementers of Industrial Policy: Bureaucrats and the Korean Export Miracle*

The question

- Formally similar industrial policies produce sharply divergent outcomes.
- Most of the debate is about design: which sectors, at what scale, with which instruments.
- This paper takes up **implementation**: the individuals charged with carrying the policy out.
- The setting is South Korea's overseas export promotion during the growth miracle.

THE IMPLEMENTERS OF INDUSTRIAL POLICY : BUREAUCRATS AND THE KOREAN EXPORT MIRACLE*

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February 25, 2026

Abstract

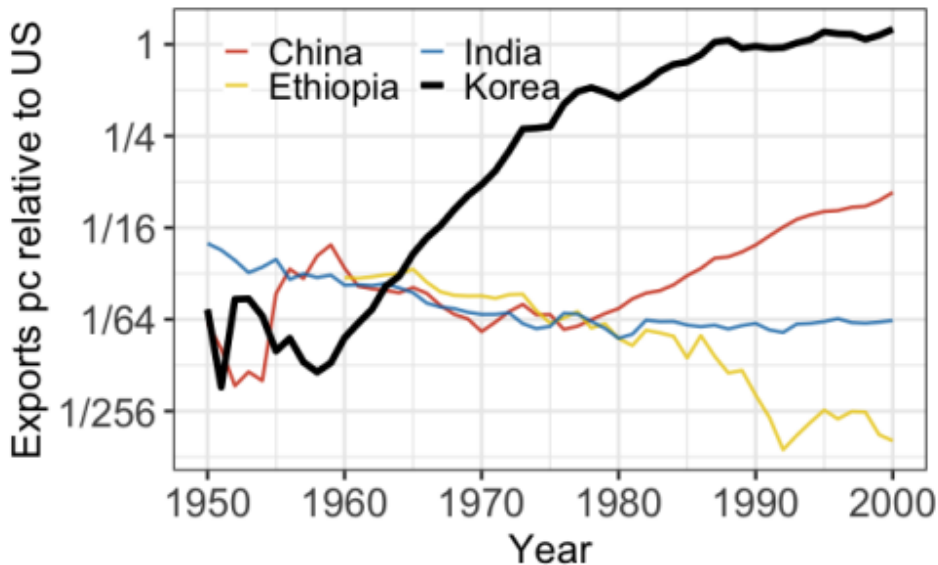
How much does the effect of industrial policy depend on the capacity of the bureaucrats implementing it? We exploit the rotation schedule of managers of South Korea's export promotion offices in 87 countries between 1965 and 2000 to show that a one standard deviation increase in bureaucrat ability boosts product-level exports by 37%, while the policy increases exports by 38% on average. Together, this implies the export promotion policy has no effect when implemented by a bureaucrat one standard deviation below average. Under higher-ability bureaucrats, South Korean exports respond more strongly to destination import demand and expand more for products prone to information frictions, suggesting more effective transmission of market information. We find that performance in the first appointment predicts whether a bureaucrat sees subsequent appointments, highlighting performance-based screening of bureaucrats as a mechanism that increases the policy's effect.

JEL classification: L52, D73, O1, F13, M50

Keywords: Industrial Policy, Bureaucracy, Economic Development, Export Promotion

The Korean setting

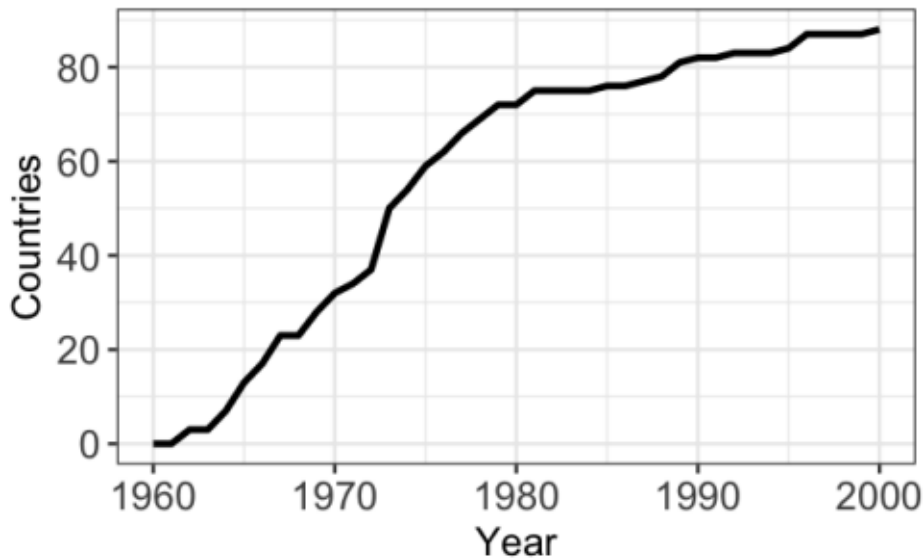
- Real GDP per capita: 1,275 dollars in 1962, 24,834 in 2001.
- Korean exports per capita were below 2 percent of the US level in the 1950s and reached parity by the 1990s.
- Two **competing accounts** of the Korean state.
- Amsden (1989): “the power of the state to discipline big business was greater in Korea”.
- Cheng et al. on the 1950s: “Under Syngman Rhee the bureaucracy was generally both ineffective and disorganised, characterised by widespread corruption and patronage”.



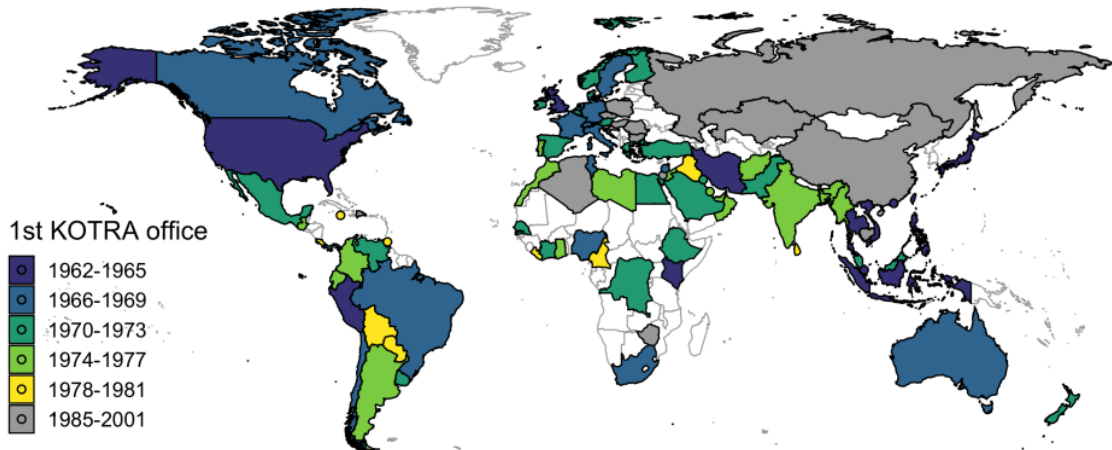
(a) Korean Exports relative to the U.S. by Year

KOTRA and its overseas offices

- **KOTRA**, South Korea's trade promotion agency, founded in 1962.
- 138 offices in 87 countries between 1965 and 2000.
- Three functions: market research, matching Korean sellers to foreign buyers, and Korean pavilions at trade fairs.
- One target per office: exports to its host country.
- Wide discretion over how to get there, so headquarters judged the outcome.



(b) Countries with Export Promotion Offices by Year

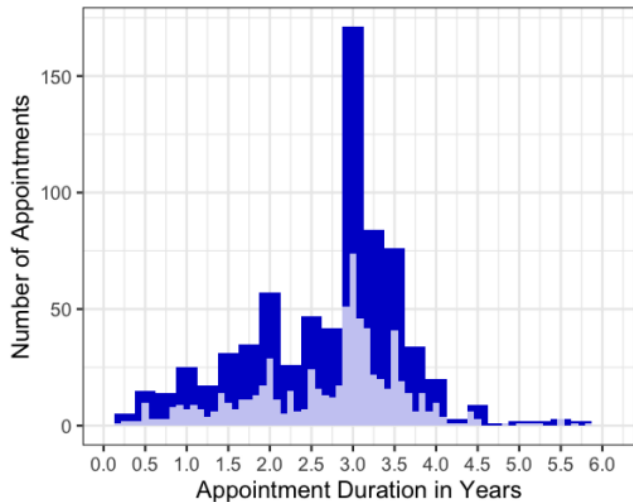


(c) Map of Export Promotion Offices

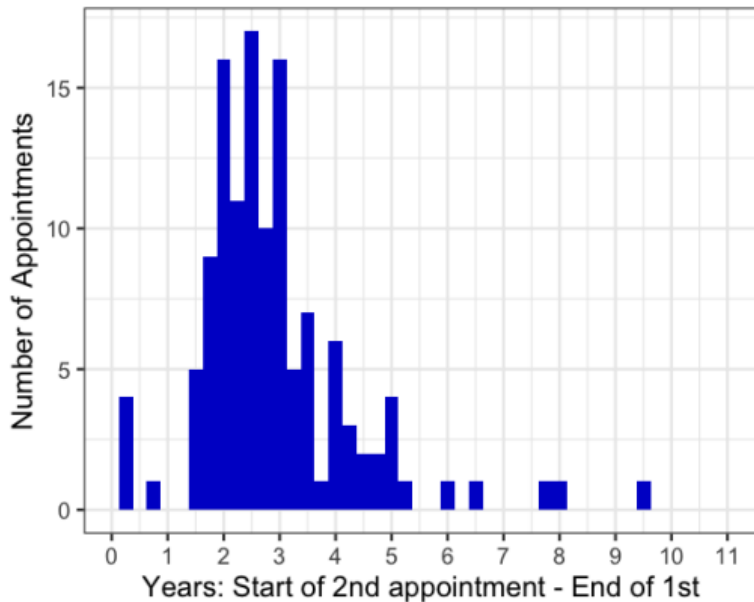
The rotation of office managers

- Each country office was headed by a manager, and managers **rotated every three years**.
- Modal and median posting: 36 months. Median gap between postings: 29 months.
- Little discretion over when a manager left an office.
- Assignments turned on language, seniority and the desirability of the destination, all largely fixed over time.

Figure A.1: Appointment Durations and Time Between Appointments



(a) Distribution of Appointment Durations. Median and Modal Duration: 36 Months.



The data

- Appointment histories of **398 managers** and 705 manager-country pairs, from newspaper archives and internal KOTRA publications.
- Bilateral exports by product and year, from 1962.
- What the offices produced: 80,000 market report entries linked to a country, 169,280 import inquiries, 1,024 trade fairs.
- Names harmonized across sources written in Chinese characters, Korean characters and romanizations.

Figure C.7: Example Import Inquiry Table

國 名	商 社 名 及 住 所	對 韓 輸 入 希 望 品 目	問 議 番 號
Australia	Healthjoy Distributors 56 Willsmere Road, Kew, Vic. 3101 Tlx : TF 8623308	Sphygmomanometers; Stethoscopes, Diagnostic set	4376-광-043 (MELD900085)
U.S.A.	Ancel Int'l Inc. 5000 Cedar Plaza Parkway Suite 304, St. Louis, MO 63128 Fax : 314-842 3222	Chair & parts	4376-광-044 (CHID900206)
Spain	SA Transgrup Castillejos, 408-412 08024 Barcelona Tel : 51246 Fax : 3- 347-4368	Sewing-machine accessories and replacement parts	4384-중-200 (MADD900149)

Notes: This figure displays a snippet of an import inquiry table from *Overseas Market News* dated March 9, 1990. The four column titles read, from left to right, country name; company name and address; products sought for import from Korea; and inquiry reference ID.

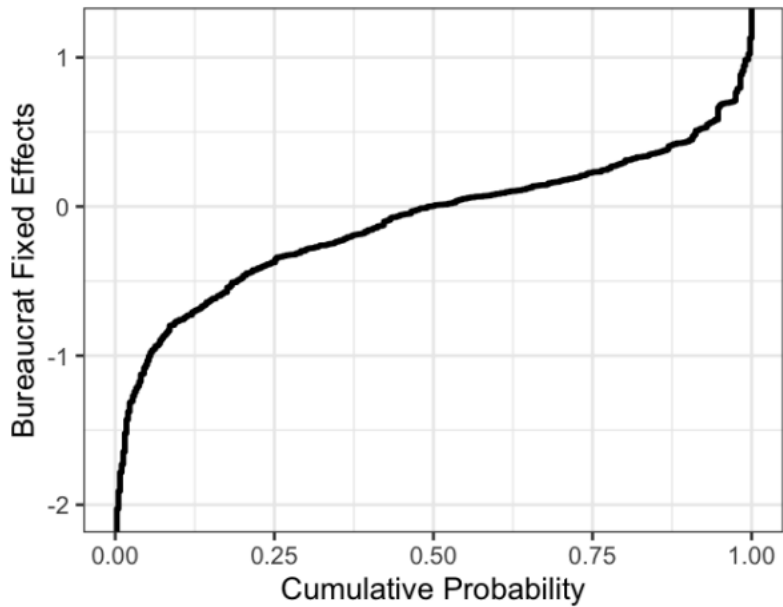
Separating the manager from the country

- Exports to the United States dwarf exports to Kenya, and that is the country, not the manager.
- Managers move between countries, and countries cycle through managers.
- Following exports when a manager arrives, and when he leaves, separates the two.
- The **rigid rotation** makes it unlikely that good managers were sent precisely to countries about to grow.

Results

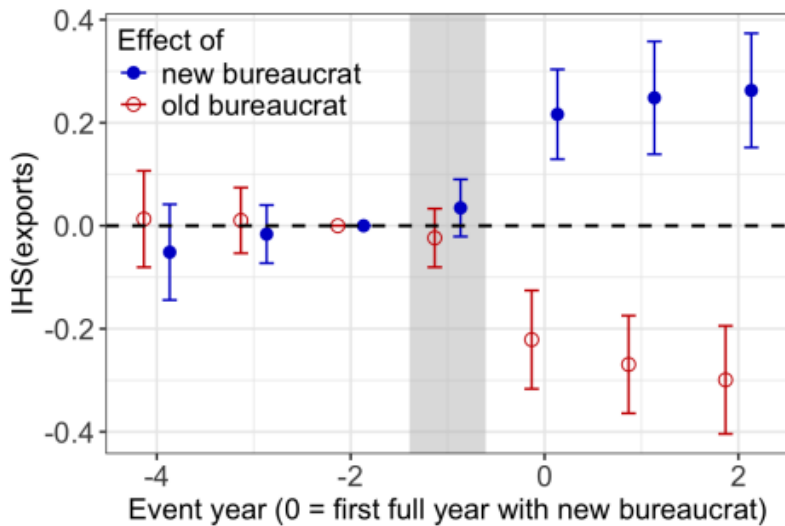
How much the manager matters

- A manager one standard deviation better raised exports to his country by about **37 percent**.
- Opening an office at all raised exports by 38 percent.
- So an office run by a manager one standard deviation below average would show no effect.
- Managers explain about a seventh of the variation that countries explain.
- Roughly the trade effect of moving the destination from London to Mumbai.

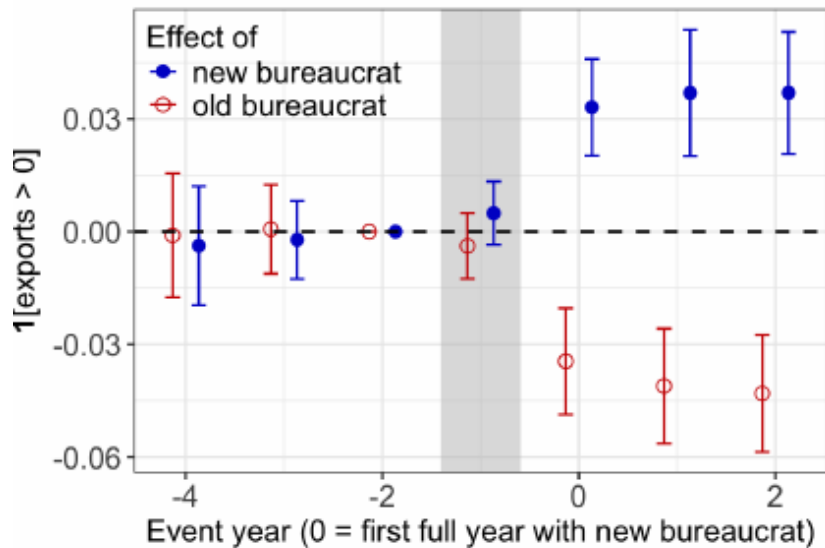


Around a change of manager

- Exports move within one or two years, up with the incoming manager's ability and down with the outgoing manager's.
- The two are **symmetric**, and there is no differential trend beforehand.
- Korean exports were growing over 35 percent a year, so a fall here means slower growth, not shrinking trade.
- Both margins move: new products start being exported, and products already sold expand.



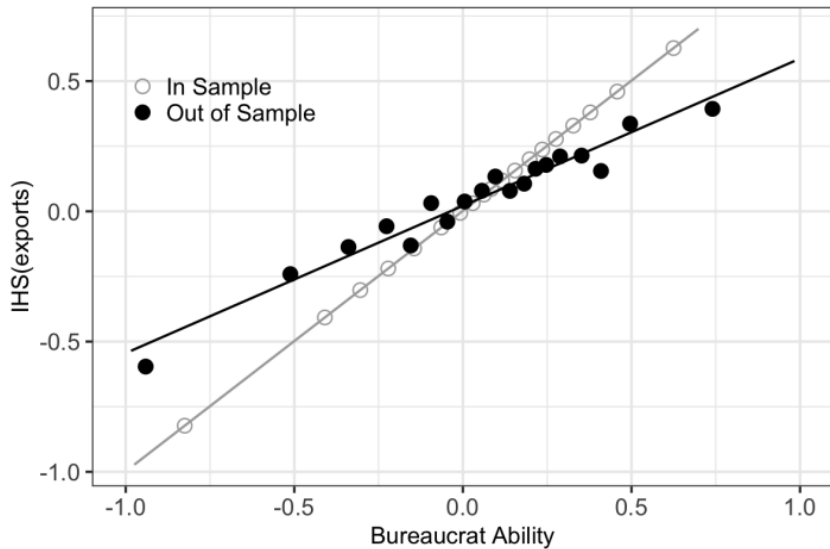
(a) Benchmark result, IHS(exports), combining extensive and intensive margin



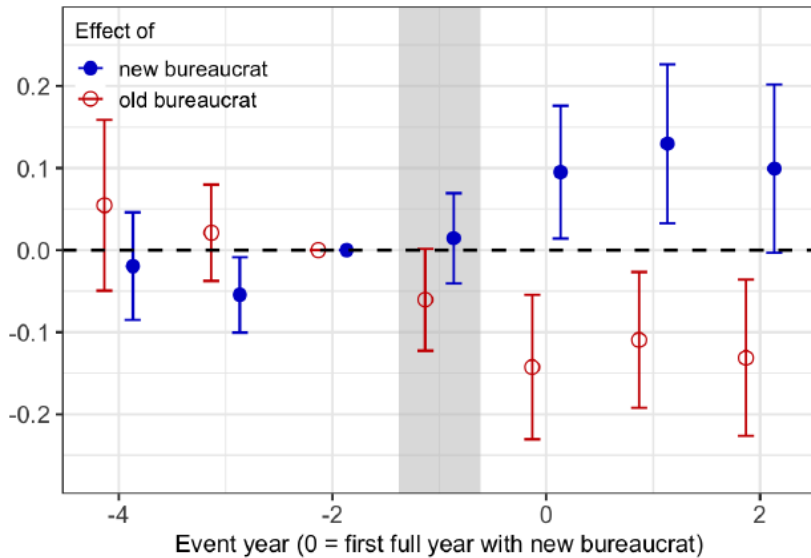
(b) Extensive margin

Is it noise?

- Reshuffling managers at random across the same appointments: the manager component goes to **zero**.
- Ability estimated leaving out a country still predicts exports in that country.
- Korean ambassadors, who also rotate but have broad diplomatic duties, account for 4 percent of the variation, against 14 percent for KOTRA managers.



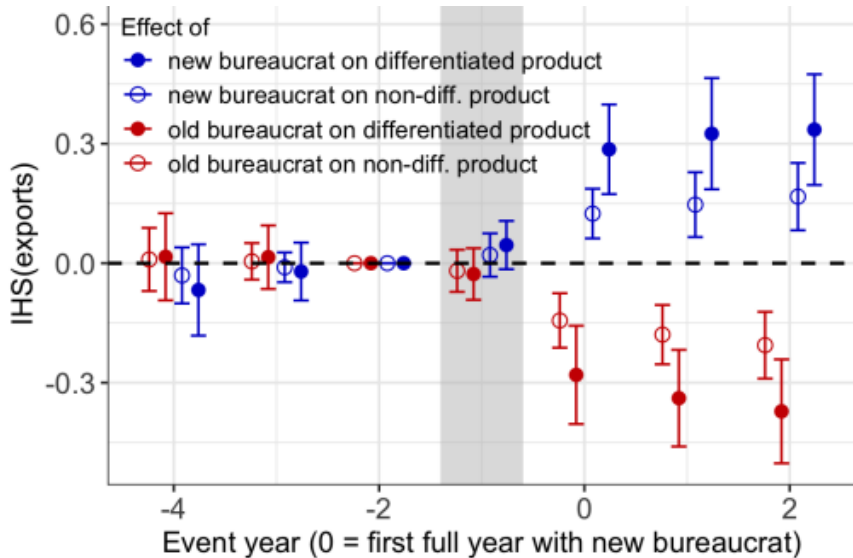
(a) Exports are greater under bureaucrats with high out-of-sample fixed effects.



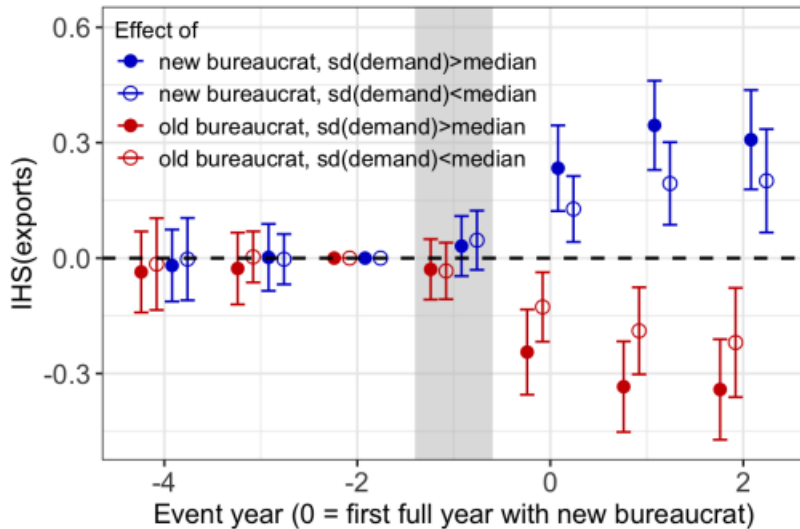
(b) Exports respond to out-of-sample fixed effects of new (incoming) and old (outgoing) bureaucrats.

What good managers do differently

- Larger effects for differentiated products and for products facing volatile demand in the destination.
- Larger effects where import demand is rising, and for heavy and chemical industry products during the **HCI drive** (1973 to 1979).
- More import inquiries listed, over a broader set of products; more general market reports; more trade fairs and higher sales at them.
- The common thread is the transmission of timely market information.



(a) Stronger effects in differentiated products



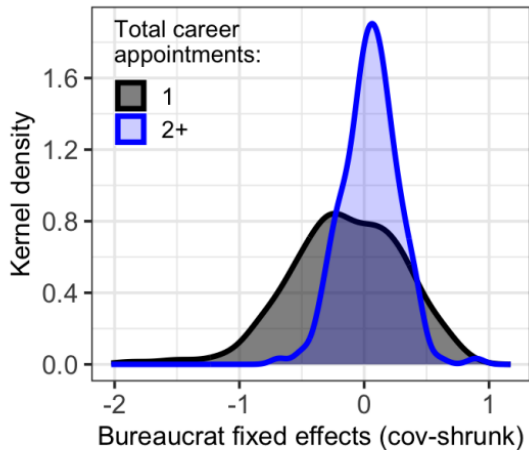
(b) Stronger effects in products with greater demand uncertainty

The organization screened its managers

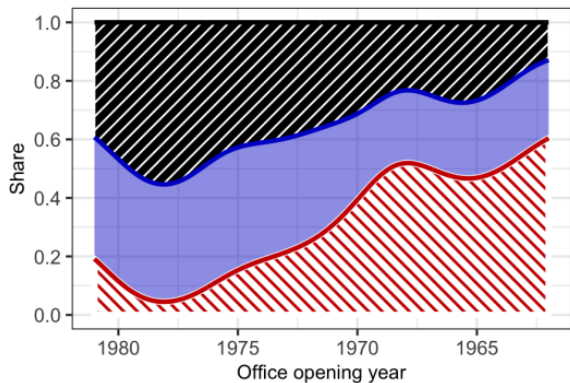
- Weak exports during a first appointment predict fewer later appointments.
- First appointments go to minor markets, third appointments to the major ones. No manager began in the United States or Japan.
- Small offices work as a testing ground: cheap places to learn who is good.
- Without that screening, exports would have been **6.5 percent lower**.
- Replacing the bottom quarter of managers with a median manager: 8.3 percent higher.



(a) Residualized exports by appointment

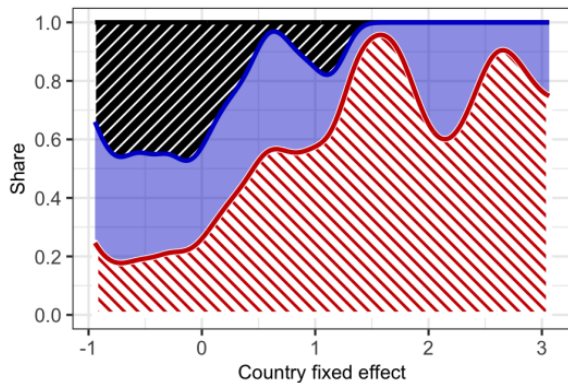


(b) Bureaucrat fixed effects



Which appointment?  1st  2nd  3rd

(c) Shares of 1st, 2nd, 3rd appointments by office opening year (in reverse order)

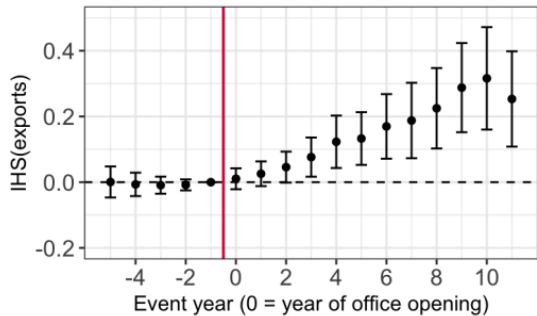


Which appointment?  1st  2nd  3rd

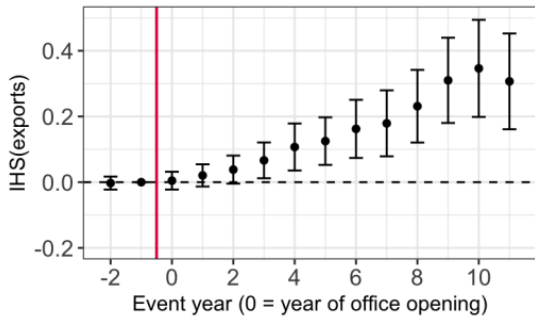
(d) Shares of 1st, 2nd, 3rd appointments by country fixed effect

The average effect of the policy

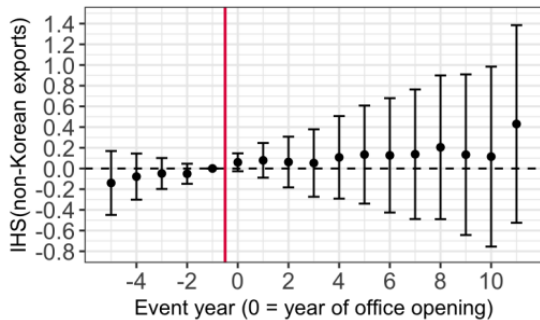
- Staggered opening of offices across countries, against countries that never received one.
- Exports rise gradually and stabilize 9 to 11 years after opening, at about **38 percent**.
- No pre-trends, and non-Korean exports to the same destinations do not move.
- Across 17 European countries, rank by opening year against rank by 1962 market size: correlation of 0.87. The UK was largest and first, Portugal smallest and last.



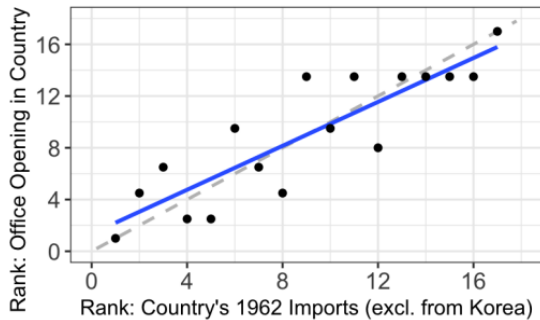
(a) Korean exports. Sample includes openings from 1967 to 1981



(b) Korean exports. Sample includes openings from 1964 to 1981



(c) Non-Korean exports. Sample includes openings from 1967 to 1981



(d) Order of office openings in Europe follows pre-determined market size

In short: Barteska and Lee

- The effect of an industrial policy depended heavily on **who** implemented it.
- A one standard deviation better manager was worth as much as having the office at all.
- The mechanism was information: matching Korean suppliers to foreign buyers.
- The agency learned about ability from first postings and reallocated its people accordingly.

Public R&D and Brazilian agriculture

Akerman, Moscona, Pellegrina and Sastry (2025), *Public R&D Meets Economic Development: Embrapa and Brazil's Agricultural Revolution*

The question

- Standard view: innovation happens in a few frontier countries, and everyone else grows by **adopting** foreign technology.
- But frontier technology in agriculture and medicine is tailored to the environments of high-income countries.
- So public R&D in locally appropriate technology may be a way out of the technology mismatch trap.
- The case is Embrapa, the public research corporation created in Brazil in 1973.

Public R&D Meets Economic Development: Embrapa and Brazil's Agricultural Revolution

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November 5, 2025

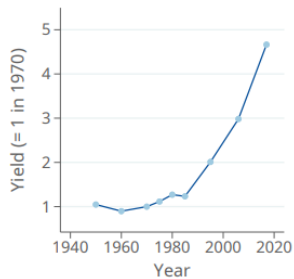
Abstract

Can public R&D investment in developing countries drive productivity growth? We study this question in the context of Brazil's agricultural revolution and the Empresa Brasileira de Pesquisa Agropecuária (Embrapa), a public research corporation established in 1973 to develop locally relevant science and technology. First, using researcher-level data compiled from the resumes of all agricultural scientists in Brazil, we show that Embrapa redirected research toward locally important staple crops and toward Brazil's particular ecological conditions, in part by sustaining productive research even in remote and research-scarce regions. Second, using municipality-level panel data from nine rounds of Brazil's agricultural census, we study how Embrapa affected agricultural production. We develop an identification strategy that exploits the staggered establishment of Embrapa's research laboratories and heterogeneous exposure to the benefits from new innovation in ecologically distinct parts of the country. We find that municipality-level exposure to Embrapa significantly increased agricultural yields, driven both by expanded input use and higher residual productivity. Consistent with effects being driven by locally appropriate innovation, exposure to Embrapa predicts larger take-up of Embrapa's new crop varieties and pronounced productivity gains only for the staple crops that were the focus of Embrapa's research. Third, combining our estimates with a model, we find that public R&D investment increased national agricultural productivity by 110% with a benefit-cost ratio of 17. Embrapa's decentralized structure, in which research labs were spread across many ecological zones instead of in a single hub, explains more than half of these gains.

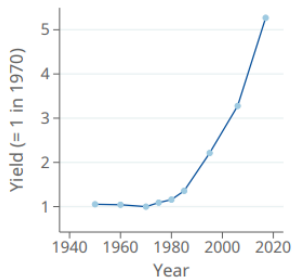
Brazilian agriculture before the 1970s

- Yields for maize, wheat and rice close to flat between 1950 and 1970.
- Already a net food importer, increasingly reliant on foreign donations and **food aid**.
- The Green Revolution did not reach Brazil: new varieties suited neither the acidic soils nor largely rainfed farming.
- Domestic research was concentrated in wealthier states and in cash crops. By 1960 more than half the non-fruit varieties from the Campinas institute were coffee, sugarcane and cotton.

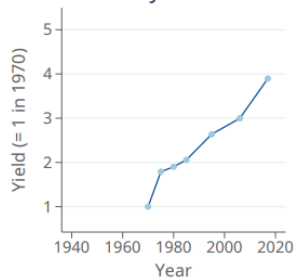
Maize



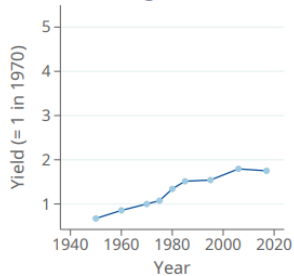
Rice



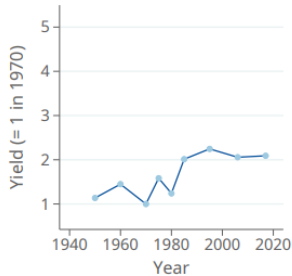
Soybeans



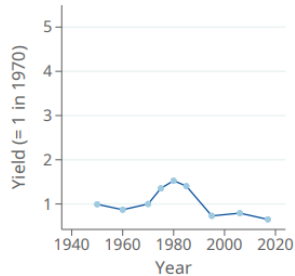
Sugarcane



Coffee



Cocoa



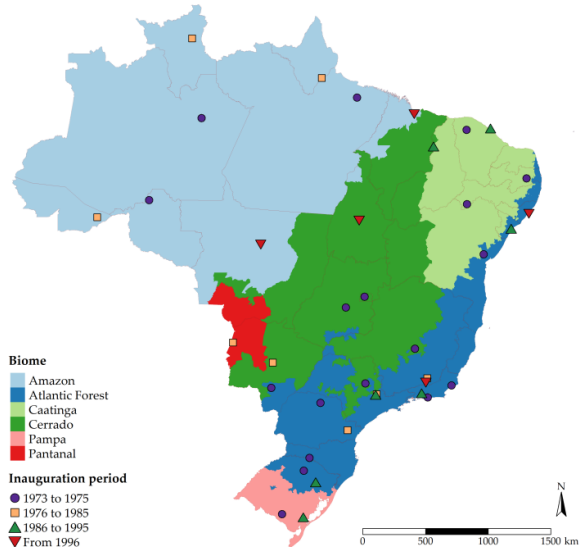
Embrapa: origins and design

- Eliseu Alves, one of its founders: “The potential existed, but there was no science capable of generating technology suited to what we needed”.
- The design group named two bottlenecks: the **geographic centralization** of research, and the lack of trained personnel and careers.
- Three organizational principles: scale, geographic scope, and the legal form of a public corporation.
- Centers opened in regions with no research tradition. Priority regions: the Northeast, the Cerrado and the Amazon.

Scale

- By the 2000s, Embrapa's spending as a share of agricultural GDP was comparable to all public agricultural R&D in the United States.
- Budget in 2010: 1.9 billion reais, about 1.15 billion dollars. The USDA research budget that year was 2.3 billion dollars.
- About **2,300 agricultural scientists** in 2010, 40 percent of all those in the country.
- Stated goal: to study every region and ecosystem of Brazil.

Figure 2: Embrapa's Research Centers



The Cerrado and soy

- The Cerrado: two million square kilometers of tropical savanna, very low soil pH, high aluminum saturation.
- Borlaug on the view in the global research community: “nobody thought these soils were ever going to be productive”.
- Embrapa Cerrados, 1975: soil chemistry and **liming**, plus bacterial nitrogen fixation. Two Embrapa researchers later received the World Food Prize for these strands.
- Embrapa Soja, also 1975, set out to tropicalize the soybean. A first variety adapted to tropical latitudes followed in 1980.
- By the 2006 census the Cerrado supplied 42 percent of national soy, against no more than 5 percent of any of these crops in the 1970s.

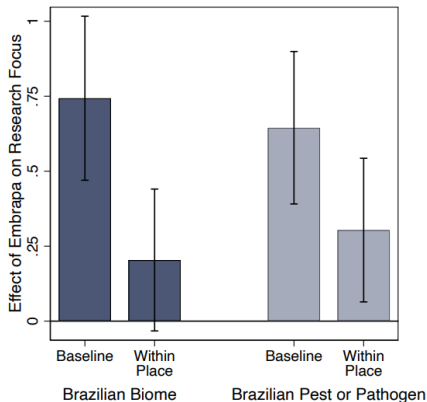
The data

- Founding year, location and annual budget of every Embrapa center, obtained by transparency request.
- The Lattes CV platform: 35,602 agricultural researchers and about 1.3 million articles, with titles coded by crop, biome and pest.
- Nine rounds of the Agricultural Census, 1960 to 2017, digitized from scans for the early years.
- Nine **agro-climatic characteristics** per municipality: temperature, precipitation, elevation, ruggedness, growing season, soil acidity and texture.

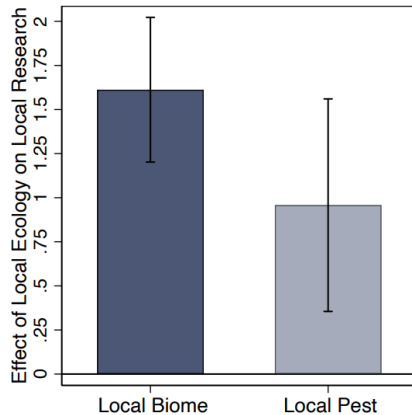
Embrapa redirected research

- Embrapa articles mention a Brazilian biome **74 percent** more often than the sample average, and a Brazilian pest or pathogen 64 percent more.
- Much of that is about where the research is done: the gap shrinks sharply once location is held fixed.
- Researchers study the biome they sit in and the pests present in their state.
- Embrapa also shifted attention to staple crops (beans, cassava, maize, rice, soy, wheat), and that shift survives holding location and the researcher fixed.

Figure 3: The Direction of Research Across Ecological Conditions

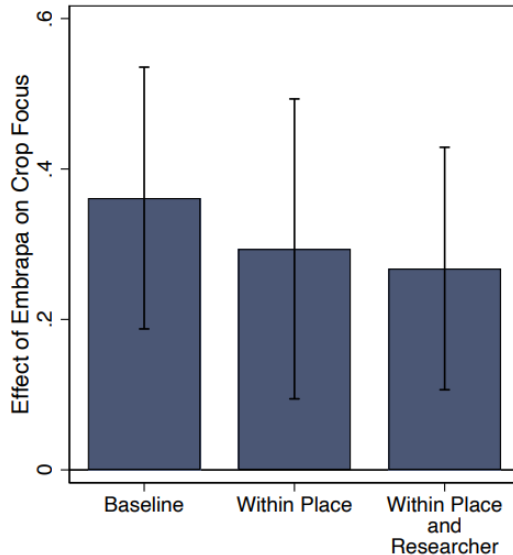


(a) Embrapa's Effect on Topic Focus ($p \times i \times t$)



(b) Local Research Focus ($i \times k$)

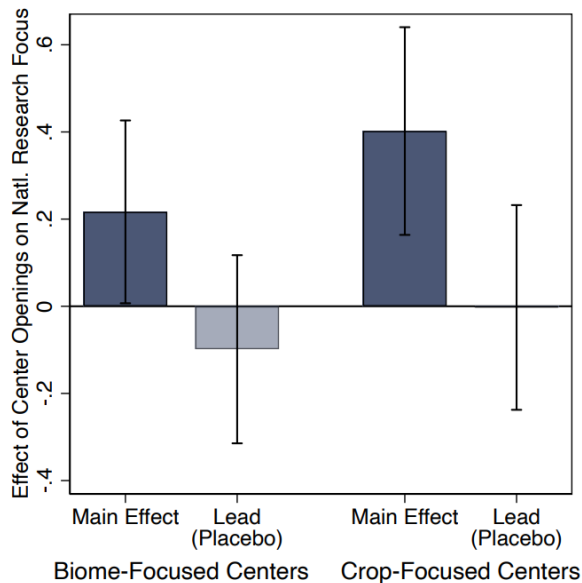
Figure 4: The Direction of Research Across Crops



No crowding out

- Embrapa might simply have absorbed research that would have happened anyway.
- Opening a center focused on a crop or a biome raised **national** research on that topic.
- Part of the increase comes from researchers not employed by Embrapa.
- No movement in the year before an opening. If anything there was crowd in.

Figure 5: Embrapa and the National Direction of Research



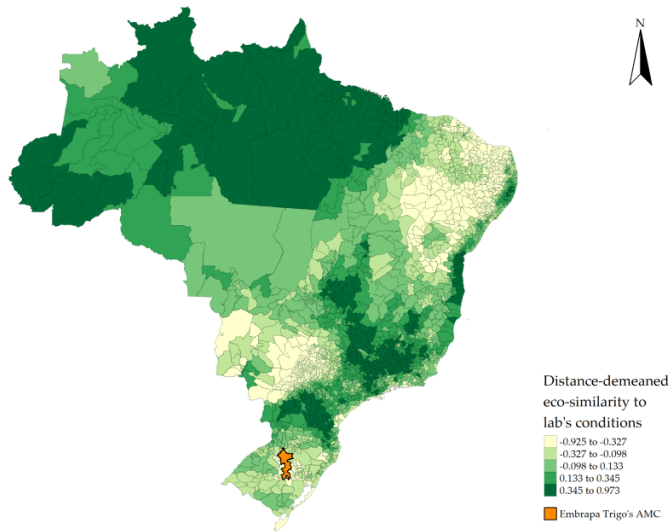
Productive research far from the hubs

- Centers went to places with little pre-existing research capacity and no agglomeration of scientists.
- Researchers move across both employers and locations, so the two effects can be separated.
- Employment at Embrapa raised research output, and by most in the **remote regions**.
- Enough to more than offset the penalty of working away from the established hubs.
- The same holds restricting to high-impact international journals.

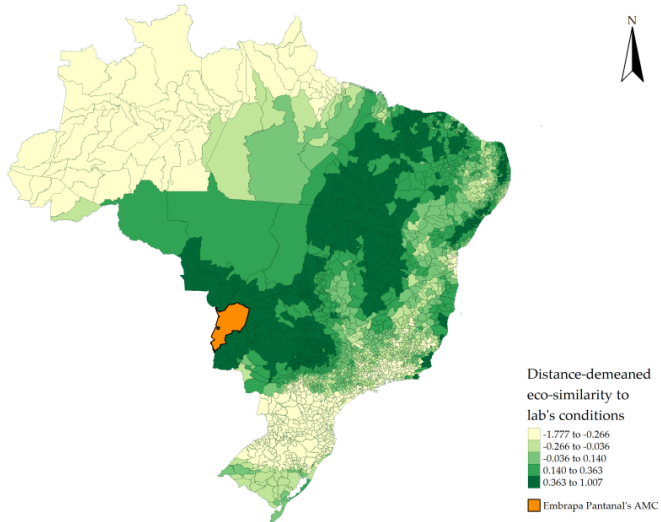
Measuring exposure to Embrapa

- Technology developed for one ecology travels to similar ecologies, not to distant ones.
- **Ecological similarity** between each municipality and each center, built from the nine agro-climatic characteristics.
- Exposure of a municipality in a given year: the highest similarity among centers open by that year.
- Exposure rises as Embrapa spreads into new ecological zones.

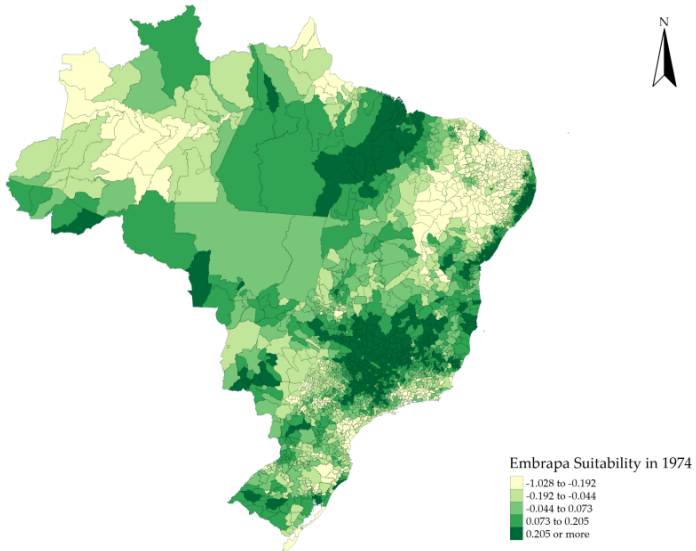
(a) Similarity to Embrapa Trigo



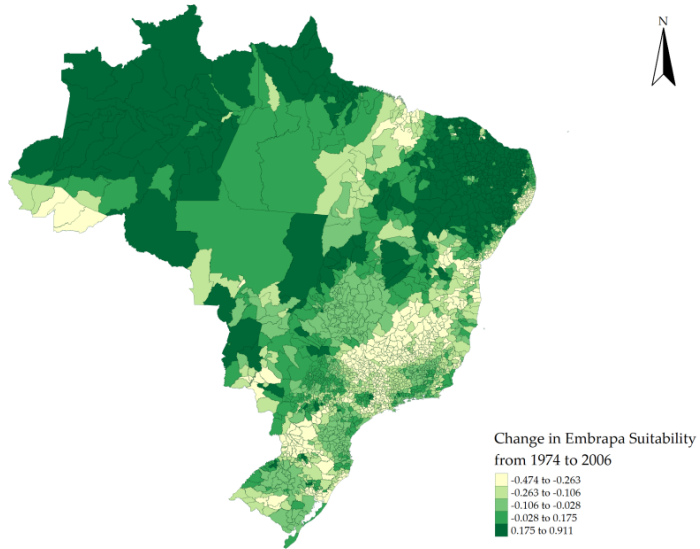
(b) Similarity to Embrapa Pantanal



(c) Embrapa Exposure in 1974



(d) Change in Exposure, 1974 to 2006



The comparison

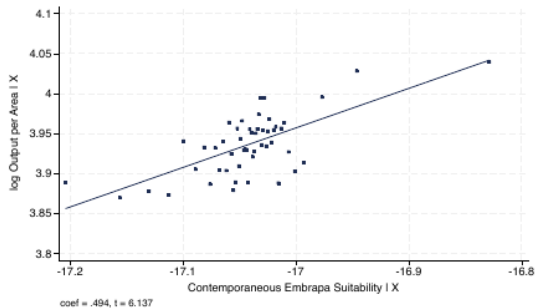
- When a center opens, compare municipalities ecologically **similar** to it with ecologically distant ones.
- Physical proximity is controlled for, and the result survives dropping municipalities within 100 km of any center.
- Assumption: similar and distant municipalities were on the same trend before the center opened.
- Placebo expansions of Embrapa across time and space put the real estimate in the right tail.

Result: higher productivity

- One standard deviation more exposure raised production value per unit of farm area by about **12 percent**.
- The same holds for land values, for crop yields, and for a broader productivity measure net of labor and inputs.
- No pre-trend over 1960 to 1970.
- The effects accumulate: the long-run effect is 30 to 40 percent larger than the within-decade effect.

Figure 7: Embrapa Affects Productivity Contemporaneously, with no Anticipation

(a) Contemporaneous Exposure



(b) Future Exposure

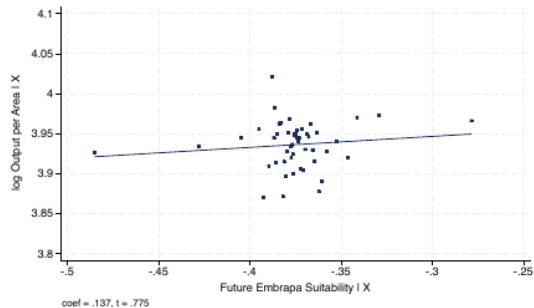
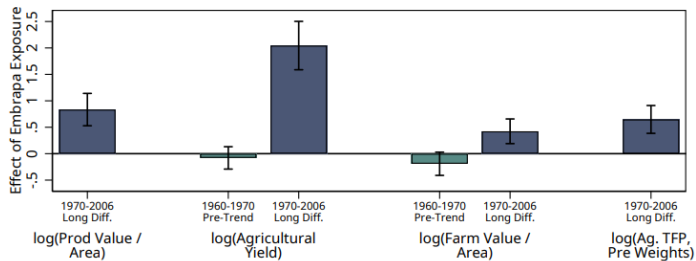


Figure 8: Long Difference Estimates and Pre-Existing Trends

(a) Unweighted



(b) Weighted by Agricultural Area

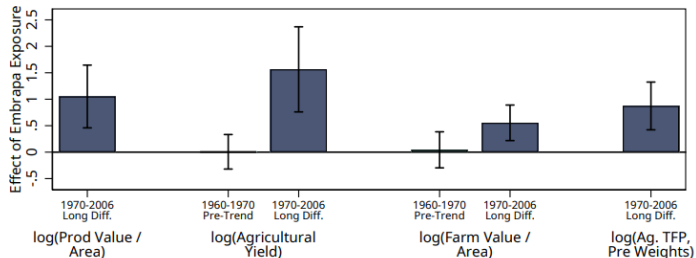
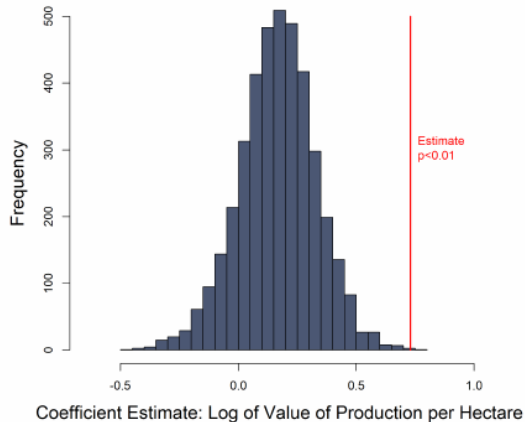
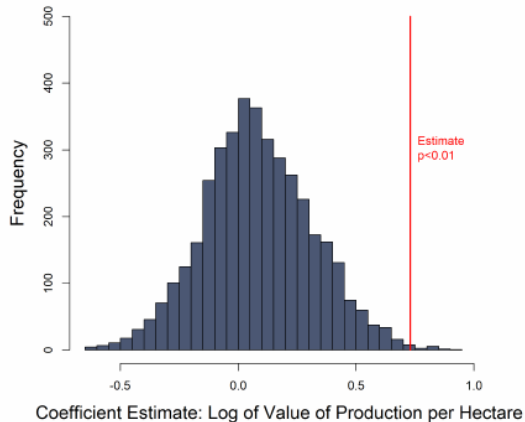


Figure 9: Falsification Tests



(a) Randomizing Across Geography and Time

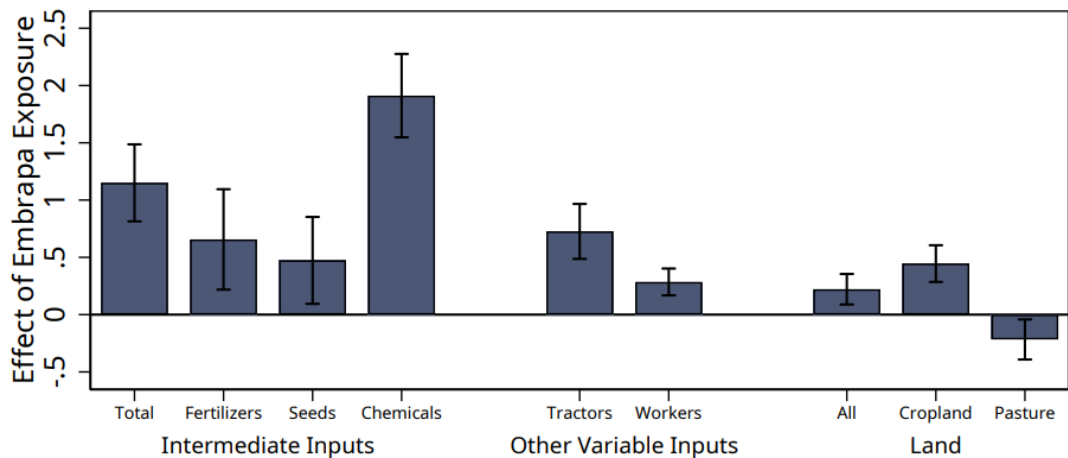


(b) Randomizing Across Time Only

Mechanisms

- Spending rises on exactly the inputs Embrapa worked on: fertilizers, seeds and chemical defenses.
- Cropland expands, over half of it converted from pasture. Labor use rises far less than input use.
- Embrapa's own **seed varieties** spread to ecologically similar municipalities even when far away: one standard deviation more similarity, about 50 percent more area planted.
- Output and yields rise for the focus crops and are flat for the rest, even within the same municipality and year.

Figure 10: Embrapa Exposure and Agricultural Inputs

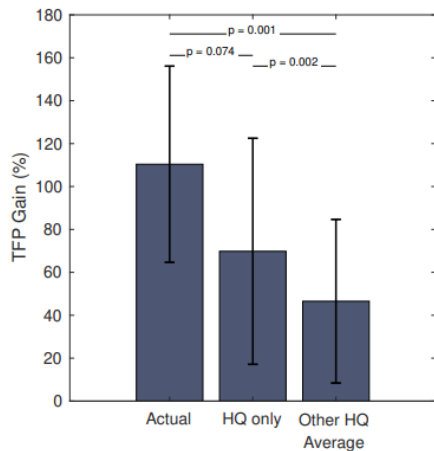


Benefits and costs

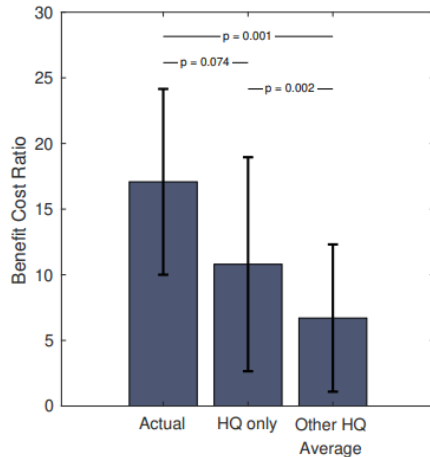
- Combining the estimates with a model: Embrapa raised aggregate agricultural productivity by **110 percent**.
- That is 39 percent of all Brazilian agricultural productivity growth since 1970.
- Benefit-cost ratio of 17, against a peak cost of about 1 percent of Brazil's agricultural GDP.
- For comparison, Jones and Summers put overall US public R&D at 5, and Griliches put hybrid corn at 7.

Figure 11: Productivity Gains and Benefit-Cost Analysis of Embrapa

(a) Productivity Gains



(b) Benefit-Cost Ratio



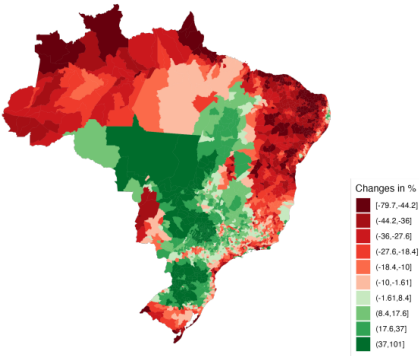
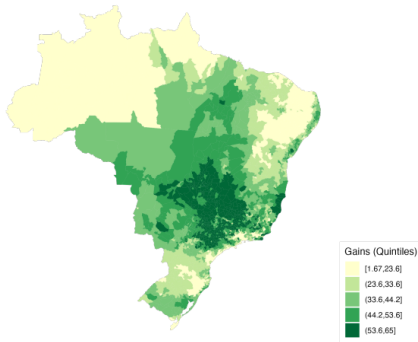
The value of spreading out

- Counterfactual: the same budget concentrated in a single center in Brasilia.
- The productivity gain falls from 110 to 70 percent, and the benefit-cost ratio from 17 to 11.
- A single center at the average alternative location achieves less than half the observed benefit.
- Brasilia is ecologically close to much of the country, so it is the best single site, which is consistent with the choice made.
- **Geographic scope** carried much of the return, over and above the level of spending.

Figure 12: Geographic Distribution of the Productivity Gains of an Embrapa Center

(a) Initial Aggregate Productivity Gain

(b) Change in Aggregate Productivity Gain



In short: Akerman and co-authors

- Public R&D aimed at local conditions raised Brazilian agricultural productivity substantially.
- The mechanism was developing and diffusing technology matched to each region's ecology.
- The return was high by the standards of the literature on public R&D.
- Where the research capacity was placed did much of the work.

Across the three papers

- Aneja and Xu: shielding rank and file bureaucrats from politics improved what the state delivered.
- Barteska and Lee: who ran the office mattered as much as opening the office.
- Akerman and co-authors: where the state placed its research capacity determined which technology got made.
- All three measure the state through an **output**: mail delivered correctly, exports, crop yields.

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- Barteska, P. and Lee, J. E. (2026). The Implementers of Industrial Policy: Bureaucrats and the Korean Export Miracle. Working paper.
- Akerman, A., Moscona, J., Pellegrina, H. S. and Sastry, K. A. (2025). Public R&D Meets Economic Development: Embrapa and Brazil's Agricultural Revolution. Working paper.